TrustSQL: A Reliability-Oriented Text-to-SQL Pipeline

Schema Awareness · Chain-of-Thought Reasoning · Verification & Self-Correction

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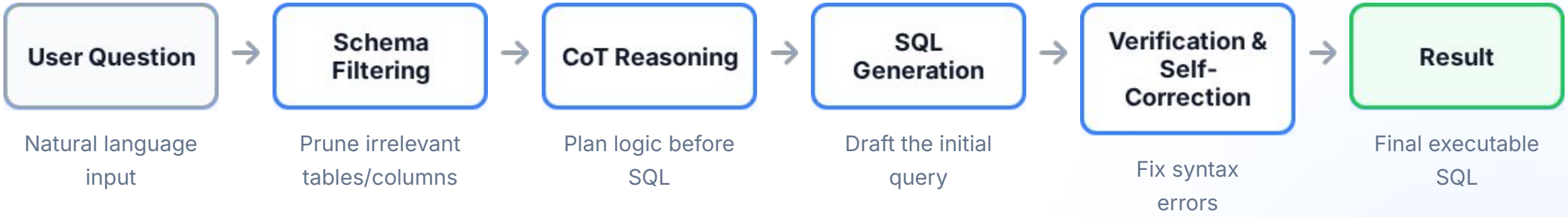
System Overview

- Background: LLMs struggle to generate correct SQL for large, noisy real-world databases.
- Our Solution: A 3-stage reliability pipeline.



Live Demo

Natural Language → SQL on BIRD Benchmark (GPT-4o-mini)

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  TrustSQL — Interactive Demo
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Available databases (11):
   1. california_schools
   2. card_games
   3. codebase_community
   4. debit_card_specializing
   5. european_football_2
   6. financial
   7. formula_1
   8. student_club
   9. superhero
  10. thrombosis_prediction
  11. toxicology
============================================================
Type 'quit' to exit.

Question: What is the phone number of the school that has the highest average sc
ore in Math?
Database ID: california_schools
Evidence (optional, press Enter to skip):

=== Step 1: Filtered Schema ===
  satscores: ['cds', 'AvgScrMath']
  schools: ['CDSCode', 'Phone']

=== Step 2: CoT Reasoning ===
1. A single value is needed (the phone number of the school).
2. The tables likely involved are the schools and satscores tables.
3. Yes, a JOIN between tables is needed to connect school information with their
 corresponding scores.
4. Yes, there is a filtering condition to identify the school with the highest a
verage score in Math.
5. Yes, aggregation is needed to calculate the average score for each school.
6. Yes, sorting or row-limiting is needed to determine the school with the highe
st average score.
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=== Step 3: Generated SQL ===
SELECT s.`Phone`
FROM schools s
JOIN satscores sc ON s.`CDSCode` = sc.`cds`
ORDER BY sc.`AvgScrMath` DESC
LIMIT 1;

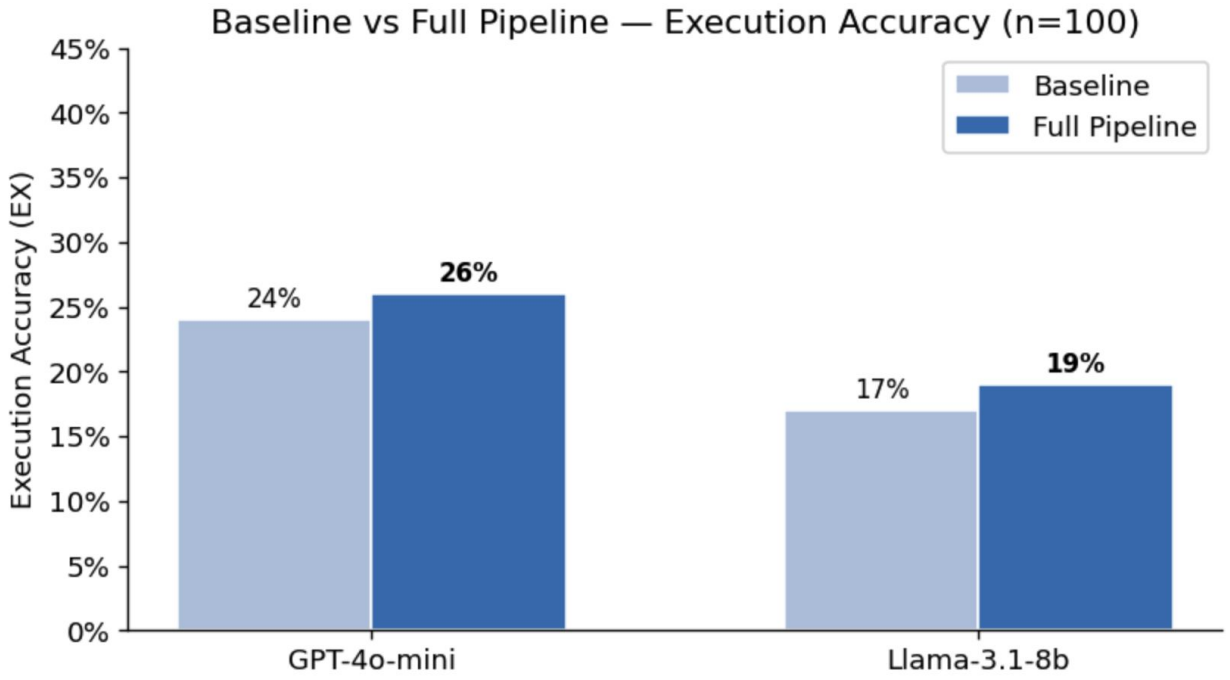
=== Step 4: Verification ===
[VALID]

=== Step 5: Execution Result ===
  Rows: [('(408) 366-7700',)]

============================================================
Final SQL:  SELECT s.`Phone`
FROM schools s
JOIN satscores sc ON s.`CDSCode` = sc.`cds`
ORDER BY sc.`AvgScrMath` DESC
LIMIT 1;
Result:     [('(408) 366-7700',)]
============================================================
```

california_schools database · 5-step pipeline

Evaluation Results (n=100)



- **GPT-4o-mini:** 24% → **26% (+2%)**; Simple difficulty: 30.5% → **35.6%**
- **Llama-3.1-8b:** 17% → **19% (+2%)**; Moderate difficulty: 0% → **11.4%**

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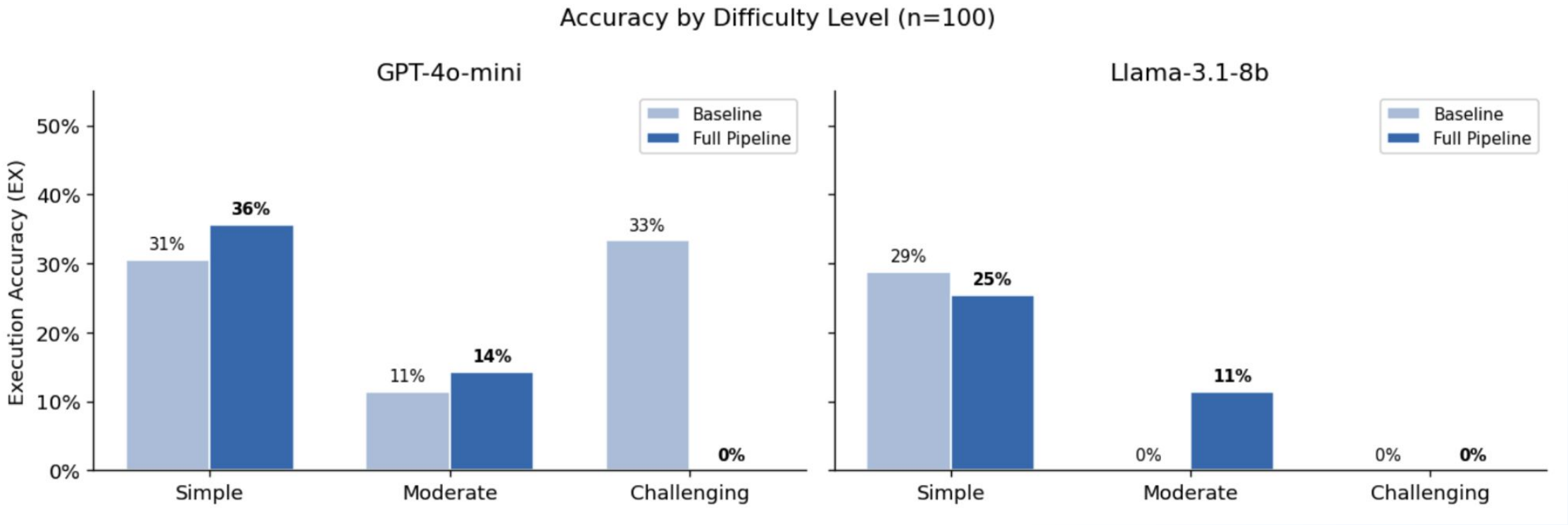

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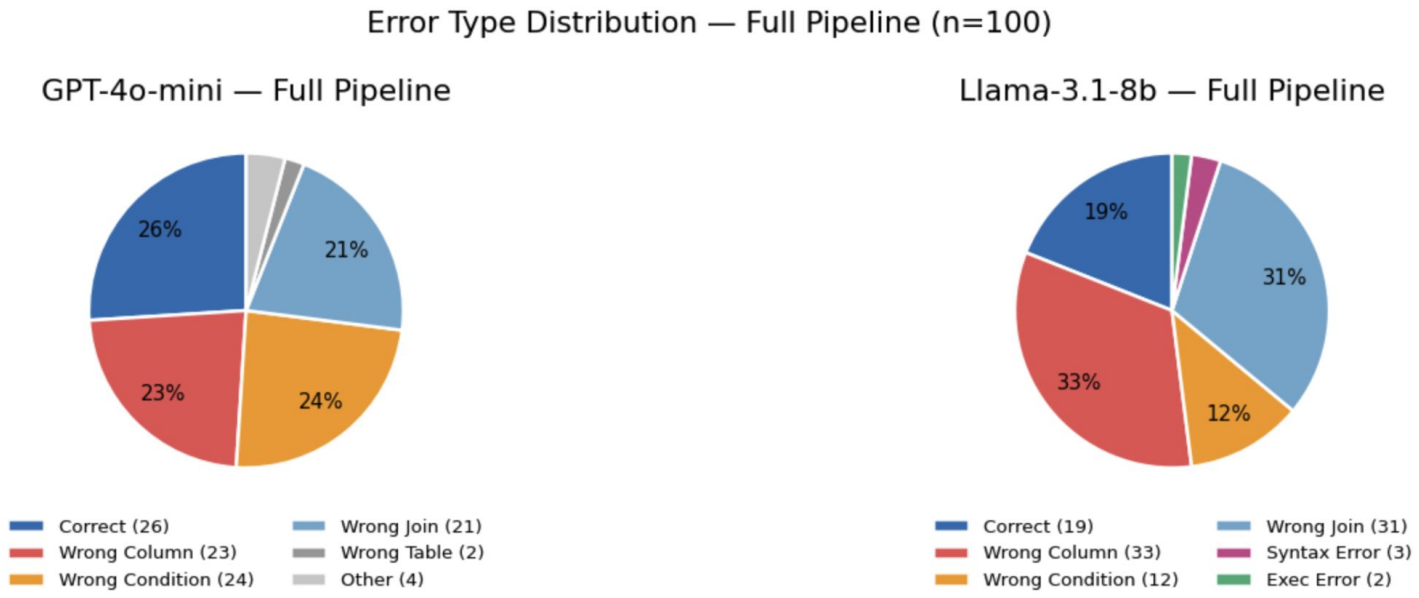
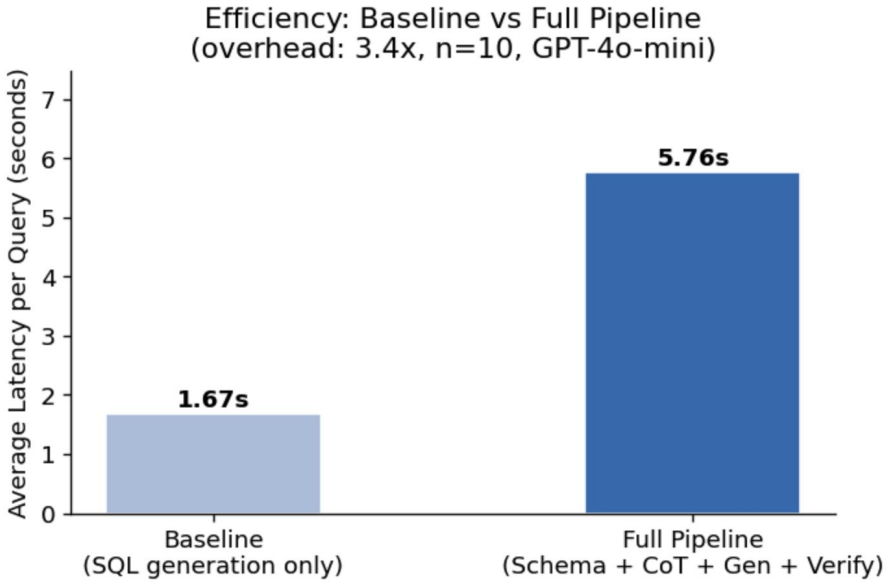
Error Analysis & Efficiency





- **Most common errors:** wrong column, wrong join, wrong condition.
- **Full Pipeline overhead:** ~3-4x latency, acceptable for the resulting accuracy gain.

Conclusion

Full Pipeline outperforms Baseline on both evaluated models.

Schema Awareness is the most impactful single mechanism.

Verification acts as a safety net — prevents SQL syntax errors without introducing side effects.

Future work: interactive CoT with user confirmation, robustness testing, and larger-scale evaluation.